

Nova Space OP

LLM API Keys Setup Guide

SWARM by nanoFarm | August 03, 2026

This guide explains how to create 5 API keys for Nova Space OP in SWARM: 2 Groq keys, 2 OpenRouter keys, and 1 Google Gemini key. Keep keys private. Do not commit them to Git or share them in chat screenshots.

1. Overview

You will create the following keys and label them clearly:

- GROQ_API_KEY (primary Groq)
- GROQ_API_KEY_2 (Groq failover / second key)
- OPENROUTER_API_KEY (primary OpenRouter)
- OPENROUTER_API_KEY_2 (OpenRouter failover / second key)
- GEMINI_API_KEY (Google Gemini)

Recommended use in SWARM Nova Space OP (when LLM is enabled): Groq for fast Think/Narrate, OpenRouter as backup router, Gemini as additional failover.

2. Groq - Create 2 API Keys

2.1 Sign up / log in

1. Open <https://console.groq.com> in your browser.
2. Sign up with email or Google, or log in if you already have an account.
3. Complete email verification if prompted.

2.2 Create primary key (GROQ_API_KEY)

1. In the left menu, open API Keys (or visit <https://console.groq.com/keys>).
2. Click Create API Key.
3. Name it: SWARM-Nova-Primary
4. Copy the key immediately (starts with gsk_). Store it in a password manager.
5. Label this key as GROQ_API_KEY for SWARM.

2.3 Create second key (GROQ_API_KEY_2)

1. Click Create API Key again.
2. Name it: SWARM-Nova-Failover
3. Copy and save as GROQ_API_KEY_2.

Note: A second key helps when the first hits rate limits. Use a different Groq account or organization if your plan allows only one key per quota bucket.

2.4 Suggested Groq models for Nova

- Text / chat: llama-3.3-70b-versatile
- Fast fallback: llama-3.1-8b-instant
- Vision (future): meta-llama/llama-4-scout-17b-16e-instruct

3. OpenRouter - Create 2 API Keys

3.1 Sign up / log in

1. Open <https://openrouter.ai>
2. Click Sign In and use Google, GitHub, or email.
3. Open Settings or Keys: <https://openrouter.ai/keys>

3.2 Add credits (recommended)

1. Go to Credits / Billing in OpenRouter.
2. Add a small balance for production testing (pay-as-you-go).

Note: Some free models exist, but paid credits improve reliability for Nova failover.

3.3 Create primary key (OPENROUTER_API_KEY)

1. Click Create Key.
2. Name: SWARM-Nova-Primary
3. Set optional spending limit if available.
4. Copy the key (starts with sk-or-). Save as OPENROUTER_API_KEY.

3.4 Create second key (OPENROUTER_API_KEY_2)

1. Create another key named SWARM-Nova-Failover.
2. Copy and save as OPENROUTER_API_KEY_2.

3.5 Suggested OpenRouter models for Nova

- General: meta-llama/llama-3.3-70b-instruct
- Economy: google/gemma-2-9b-it
- Base URL for OpenAI-compatible clients: <https://openrouter.ai/api/v1>

4. Google Gemini - Create 1 API Key

4.1 Google AI Studio

1. Open <https://aistudio.google.com/apikey>
2. Sign in with your Google account.
3. Accept terms if prompted.

4.2 Create API key (GEMINI_API_KEY)

1. Click Create API key.
2. Choose an existing Google Cloud project or create a new one.
3. Copy the key immediately. Save as GEMINI_API_KEY.
4. Restrict the key in Google Cloud Console (API restrictions: Generative Language API only).

4.3 Suggested Gemini models for Nova

- Fast: gemini-2.0-flash
- Higher quality: gemini-1.5-pro
- Gemini uses Google AI API, not OpenAI-compatible URL by default.

5. Where Keys Go in SWARM (for your engineer)

When you share keys, they will be set as environment variables - never in source code.

Local development (PowerShell)

```
$env:GROQ_API_KEY='gsk...'
$env:GROQ_API_KEY_2='gsk...'
$env:OPENROUTER_API_KEY='sk-or-...'
$env:OPENROUTER_API_KEY_2='sk-or-...'
$env:GEMINI_API_KEY='AIza...'
```

Railway / production

Add the same variable names in Railway project Settings > Variables for the backend service.

- GROQ_API_KEY
- GROQ_API_KEY_2
- OPENROUTER_API_KEY
- OPENROUTER_API_KEY_2
- GEMINI_API_KEY
- NOVA_LLM_BASE_URL (e.g. <https://api.groq.com/openai/v1> for Groq primary)
- NOVA_LLM_MODEL (e.g. llama-3.3-70b-versatile)

6. Security checklist

- Never commit .env files or paste keys into GitHub.
- Rotate keys if exposed.
- Use separate keys for dev vs production.
- Set spending limits on OpenRouter and monitor Groq usage dashboards.
- Restrict Gemini keys to Generative Language API only.

7. What to send later

When ready, provide your engineer these 5 values securely (password manager share, not public chat):

1. GROQ_API_KEY
2. GROQ_API_KEY_2
3. OPENROUTER_API_KEY
4. OPENROUTER_API_KEY_2
5. GEMINI_API_KEY

Note: After keys are received, SWARM will wire them into Nova Think/Narrate with provider failover. Until then, Nova Space OP continues to work with deterministic (non-LLM) answers.